

BUSINESS CASE

Courier Services Management and Tracking Platform

[Working title — no branded product name has been assigned yet]

Executive Summary

The organisation currently receives and manages a growing volume of incoming courier packages using Microsoft SharePoint together with a largely manual administrative process. Staff log every parcel by hand, search for a specific parcel manually whenever a customer arrives to collect it, and work out the total income earned from collected packages by manually paging through receipts week by week and month by month. As parcel volumes continue to grow, this approach is becoming slower, more prone to error, and increasingly difficult to report on with confidence.

This document presents the business case for a Courier Services Management and Tracking Platform (referred to throughout as "the Platform") — a purpose-built web application that would replace the current SharePoint-based workflow. The Platform would allow staff to log incoming packages quickly, including via barcode scanning, automatically notify customers by SMS or WhatsApp once their package is ready for collection, give staff a live dashboard of packages collected versus still awaiting collection, and generate accurate income reports on demand instead of through manual reconciliation.

Two solutions are compared in this document: continuing with the current SharePoint-based system, and building the Platform. Based on the feasibility scoring presented in this document, building the Platform is the recommended solution, as it directly addresses the operational bottlenecks present in the current process while remaining realistically achievable within the resource and time constraints of a project of this scope.

The Platform is deliberately scoped around operational efficiency rather than customer-facing functionality. Customers require only a single automated touchpoint — an SMS or WhatsApp notification once their package is ready for collection — which keeps the build small enough to deliver a working Minimum Viable Product (MVP) within the available project timeline, while still resolving every operational pain point identified in the current process.

Problem Analysis

Business Problem

The organisation currently relies on a combination of Microsoft SharePoint and manual administrative effort to run its package collection service. While SharePoint fulfils a basic record-keeping function, it was never designed for this specific operational workflow, and several persistent issues have emerged as a result:

- Slow parcel logging – Each incoming package is captured manually into SharePoint, a process that takes staff time in direct proportion to the number of packages received, with no shortcuts available for high-volume periods.
- Slow parcel retrieval – When a customer arrives to collect a package, staff must manually search SharePoint (and in some cases the physical storage area) to locate it, causing delays and queues during busy periods.
- Manual income reconciliation – Working out total income collected per week or per month currently requires staff to go through receipts one at a time and total them by hand. This is slow, disrupts other duties, and carries a real risk of arithmetic or transcription error.
- No automatic collection notifications – Customers are not proactively informed when a package becomes available for collection. This results in customers either arriving before their package is ready (wasted trips and unnecessary front-desk queries) or leaving packages uncollected for extended periods, which contributes to storage congestion.
- No live visibility of throughput – Staff and management currently have no quick way to see how many packages are awaiting collection versus already collected at any given point in time, which makes it difficult to plan staffing and storage capacity around demand.
- Inconsistent record quality – Because SharePoint entry is manual and free-form, there is no enforced structure to the data captured, which makes historical reporting and trend analysis difficult.

NOTE FOR LECTURER/STUDENTS: *Please confirm current parcel volumes (per day/week/month) and roughly how long each of the manual tasks above currently takes staff to complete. Even rough estimates will allow a believable before-and-after time/cost comparison to be built into the feasibility study's cost-benefit analysis.*

Business Opportunities

The challenges outlined above also present a clear opportunity to reimagine how packages are managed, creating long-term value for both the organisation and the customers it serves:

- Purpose-built efficiency – A system built specifically around this workflow can meaningfully cut the administrative time spent per package by replacing manual search and manual entry with barcode-based logging and lookup.
- Automated customer communication – Automated SMS/WhatsApp notifications remove the need for staff to individually contact customers, while also reducing wasted customer trips to the collection point.
- Accurate, on-demand reporting – Automatic income calculation removes the need for manual receipt reconciliation entirely, reducing both staff time and the risk of reporting errors.
- Operational visibility – A live dashboard showing packages collected versus still awaiting collection gives staff and management a level of day-to-day planning insight that does not currently exist.
- A realistic scope for the resources available – Because customer-facing functionality is minimal by design (a single automated notification, rather than a full customer portal), the project remains small enough to be delivered as a focused, achievable build rather than an open-ended system.
- A foundation for future growth – A well-structured Platform, even at MVP stage, creates a foundation that can later be extended (for example, additional report types, multiple collection points, or new notification channels) without requiring a full redesign.

Business Requirements

Business Drivers

A number of factors must be considered in deciding the way forward for package management:

- Operational efficiency – Reduce the administrative time spent logging, locating, and reporting on packages, freeing staff time for other duties.
- Real-time visibility – Give staff a live view of how many packages are awaiting collection versus already collected, to support day-to-day planning.
- Accurate financial reporting – Automate the flat-rate income calculation so that weekly and monthly totals are available on demand instead of through manual reconciliation.
- Improved customer experience – Automatically notify customers by SMS or WhatsApp once their package is ready for collection, without requiring staff to contact each customer individually.

Business Requirements

The business problems and opportunities identified above translate into the following system requirements:

Business Problem / Opportunity	Project System Requirement
Manual parcel logging is slow and inconsistent	The system must allow a package to be logged in a single, short workflow, ideally via barcode/QR scan rather than full manual data entry.
Staff cannot quickly locate a specific parcel	The system must support fast lookup by tracking number, customer name, or customer phone number.
Income reconciliation is manual and error-prone	The system must automatically calculate total income collected over any selected date range, based on a flat collection fee applied per collected package.
Customers are not proactively notified	The system must automatically send an SMS or WhatsApp message to the customer once a package's status changes to 'ready for collection'.
No visibility of parcel throughput	The system must provide a staff dashboard showing, at minimum, the number of packages collected versus uncollected over a selected period.
Physical scanning workflow is undefined	The system must support barcode/QR scanning via either a dedicated handheld/desktop scanner or a device webcam, without requiring a full workflow

	redesign if the hardware choice changes later.
Customer and package data must be handled responsibly	The system must store personal information (customer name, phone number) in line with POPIA requirements, with access restricted to authorised staff and a defined data retention period.

Solution 1: The Courier Services Management and Tracking Platform

Description

The Platform is a web application that brings the core package-handling workflow into a single system: logging a package on arrival (including via barcode scan), tracking its status and storage location, automatically notifying the customer once it is ready for collection, and generating income reports without manual reconciliation. The system would be built around the organisation's own process and terminology rather than forcing staff to adapt to a generic, off-the-shelf tool. Development would follow a staged approach, starting with the core intake-lookup-report workflow and adding the notification and dashboard modules once the foundation is stable, allowing the project to remain realistic within the available timeline while still targeting a professional standard of finish.

Drawbacks

Building the Platform is not without its challenges. It requires upfront development and testing time before it can fully replace the current process, and it introduces a new ongoing operating cost — principally SMS/WhatsApp messaging and hosting — that the current system does not carry. Barcode scanning, whether via a dedicated scanner or a device webcam, needs to be proven reliable in testing before staff can depend on it day-to-day. Staff will also require a short period of retraining, even though the new workflow is designed to be simpler than the process it replaces.

- Requires development and testing time before the current process can be fully retired.
- Introduces an ongoing operating cost (hosting and per-message notification fees) not present in the current system.
- Barcode scanning reliability must be validated through testing before staff can depend on it.
- Staff will need a short period of retraining on the new workflow.

Benefits

The Platform offers a high degree of fit to the organisation's actual workflow, since it is purpose-built around package intake, storage, notification and reporting rather than adapted from a generic list-management tool. This directly addresses the operational bottlenecks identified in the problem analysis: barcode-based logging and lookup materially reduce staff time per package compared to manual SharePoint entry and manual searching, while automatic income calculation removes the need for manual receipt reconciliation altogether.

Beyond the immediate efficiency gains, the Platform also improves the customer experience by removing wasted collection trips through automatic ready-for-collection notifications, and gives staff and management a live operational view — packages collected versus still awaiting collection — that does not exist in the current process. Because the system is owned outright rather than licensed from a vendor, it can also be extended over time (for example, additional report types or collection points) without being constrained by a third party's product roadmap.

Feasibility Scores

Criterion	Score (1-5)	Explanation
User Friendly	4	The scan-confirm workflow for logging and finding packages is simpler and more direct than the current manual SharePoint entry and search process.
Budget Consciousness	4	Development cost is low as no software licensing is required; the main ongoing cost is hosting and per-message notification fees.
Implementation	2	Intake, notification, dashboard and reporting modules all need to be designed and built from the ground up within a limited timeframe.
Time Consciousness	3	A functional MVP covering the core workflow is achievable within the available project timeline, though the full feature set will take longer to complete.
Adaptability	4	The modular design allows later additions — for example

		further report types, additional collection points, or extra notification channels — without redesigning the core system.
End User Instruction	4	The workflow is simpler than the current process, so a short, focused training session should be sufficient for most staff.
Human Resources	3	Delivered by a small development team; this requires coordination and mentorship but does not require external hires or vendor staff.

Risks

Time Consciousness

The development team will need to balance building the Platform against other competing commitments. If timelines are not realistic, there is a risk that features are rushed or delivered incomplete, which could undermine confidence in the system at launch. Careful, staged scope management (an MVP first, enhancements later) is the primary mitigation for this risk.

Implementation

Rolling the Platform out to replace an established, familiar process will not be entirely smooth. Existing open package records will need to be migrated from SharePoint without loss, staff will need training before go-live, and any early technical issues during rollout could frustrate users and slow adoption if not managed carefully.

Notification Cost and Volume

SMS and WhatsApp messaging costs scale directly with the number of packages processed. If actual parcel volume is significantly higher than assumed at planning stage, ongoing messaging costs could exceed the original budget, making an accurate volume estimate important before final sign-off.

Solution 2: Keep Current System

Description

This option continues using SharePoint lists for parcel logging, with staff manually searching for packages when customers arrive to collect them, and manually reconciling receipts to calculate income totals. No new software would be built or purchased under this option, and the day-to-day process would remain exactly as it is today.

Drawbacks

Keeping the current system in place means that the inefficiencies identified in the problem analysis remain part of daily operations indefinitely. Staff will continue to spend time on manual tasks that could otherwise be automated, and this time cost will grow — not shrink — as parcel volumes increase. The lack of real-time visibility into throughput will continue to make staffing and storage planning more difficult than it needs to be, and the absence of automatic customer notifications will continue to generate wasted collection trips and avoidable front-desk queries.

- Administrative time per package remains high and scales directly with volume.
- Income reporting stays manual, slow, and carries an ongoing risk of arithmetic or transcription error.
- Customers continue to receive no automatic notification when their package is ready.
- Staff have no live view of how many packages are currently outstanding at any given time.

Benefits

Retaining the current system is the least disruptive option in the short term. Staff are already familiar with the tools and process, so there is no need for large-scale retraining or an adjustment period, and day-to-day operations would continue without the interruptions that typically accompany a new system rollout. This option also avoids any upfront development cost or new ongoing operating expense, which can make it appear to be the most budget-friendly choice, at least in the short term.

- No development cost or delivery risk, since the process already works, however imperfectly.
- No retraining required, as staff already know the current tools.
- No new ongoing software or messaging costs are introduced.

Feasibility Scores

Criterion	Score (1-5)	Explanation
User Friendly	3	Staff already know how to use the current tools, but manual searching and reconciliation remain slow and effort-intensive.
Budget Consciousness	4	No new spend is required in the short term, since the system is already in place and already paid for.
Implementation	5	There is nothing to implement, as the process is already in use today.
Time Consciousness	5	No implementation time is required, since no new system needs to be built or rolled out.
Adaptability	1	SharePoint lists were not designed for this workflow and are difficult to extend with the automation this project requires.
End User Instruction	5	No new training is required, as no new functionality is being introduced.
Human Resources	2	Ongoing manual effort continues to consume staff time on every single package processed, with no reduction over time.

Risks

Operational Inefficiency

Manual processes and workarounds may seem manageable at current volumes, but they will become an increasing bottleneck as parcel volumes grow. Over time this is likely to lead to longer queues, slower service, and greater frustration among both staff and customers.

Data Handling and Compliance

Ad-hoc manual processes make it harder to enforce consistent data-handling practices for customer information such as names and phone numbers. Without a structured system, ensuring POPIA-aligned access control and retention practices becomes more difficult to demonstrate and maintain over time.

Reporting Accuracy

Manual income reconciliation carries an ongoing risk of arithmetic or transcription error. As parcel volume grows, so does the time required for reconciliation and the corresponding risk of mistakes going unnoticed.

Feasibility Ranking

Solution Ranking Criteria

The feasibility ranking is based on criteria that reflect both the practical and strategic needs of the organisation, looking at how well each solution meets user expectations, fits within budget and time limits, and can realistically be implemented and maintained.

Criterion	Description
User Friendly	How easy the system is for staff to learn and use, and how quickly tasks can be completed without confusion.
Budget Consciousness	How well the option fits within the available budget, considering both upfront and ongoing costs.
Implementation	How difficult or straightforward it will be to put the system in place, including technical setup and any required process change.
Time Consciousness	How long it will take to get the system fully operational, and the impact of that timeline on day-to-day operations.
Adaptability	How well the system can be adjusted or extended to meet future needs.
End User Instruction	How much training staff will need to become confident and productive using the system.
Human Resources	The amount of staff/developer time needed to build, roll out, and maintain the system on an ongoing basis.

Feasibility Ranking Results

Each solution is scored from 1 (poor fit) to 5 (strong fit) against the criteria above.

Criterion	The Courier Services Management and Tracking Platform	Keep Current System
User Friendly	4	3
Budget Consciousness	4	4
Implementation	2	5
Time Consciousness	3	5
Adaptability	4	1
End User Instruction	4	5
Human Resources	3	2
Total	24	25

NOTE FOR LECTURER/STUDENTS: *These scores are illustrative and follow the same weighting approach as a typical worked example. Students should be encouraged to justify their own scores through discussion with mailroom/courier-point staff rather than copying these directly — a system that scores marginally lower on short-term ease can still be the stronger overall recommendation if it resolves the actual business problem, and the written justification should make that reasoning explicit.*

Recommended Solution

Despite the current system scoring competitively on short-term ease and immediate cost, it does not resolve any of the core business problems identified in this document: slow logging, slow retrieval, manual income reconciliation, and the absence of customer notifications. The Courier Services Management and Tracking Platform is the recommended solution because it directly targets each of these issues, and its drawbacks — development time and a modest ongoing operating cost — are manageable within a properly scoped project.

Implementation Note: Barcode Scanning Method

The Platform's intake workflow can be built to support either of the following approaches to barcode/QR scanning, and this decision should be made early in the project, as it affects both hardware cost and part of the frontend build:

- Dedicated barcode scanner – a handheld or desktop USB/Bluetooth scanner connected to the staff workstation. This is fast and reliable, but requires purchasing hardware for each workstation that needs it.
- Webcam-based scanning – the web application uses the workstation's existing camera, via a browser-based barcode-reading library, to scan the barcode directly. This avoids additional hardware cost, at the expense of being somewhat slower and more dependent on lighting and camera quality.

A practical middle path is to build for webcam-based scanning first, since it requires no additional hardware and is faster to implement, while keeping the option open to add dedicated scanner support later. Most dedicated scanners emulate keyboard input, which means they typically require very little additional integration work once the core intake workflow exists.

NOTE FOR LECTURER/STUDENTS: *Please confirm whether barcode scanners are already owned, and whether staff workstations have webcams — this determines which option should be recommended as the MVP default.*

Project Initiation

The project begins once this business case and the accompanying feasibility study have been reviewed and approved. This is followed by assembling the project team and assigning members to their respective core areas: front-end, back-end, and quality assurance/documentation. The first team meeting will confirm the requirements for the Minimum Viable Product, including the software stack and database design to be used.

Role	Team Member(s)
Project Manager	[Name]
Front-End Team	[Names]
Back-End & DevOps Team	[Names]
Documentation & Quality Assurance	[Names]

Project Planning

The project is structured into five phases:

1. Assembling the Team and Planning

Team members are assigned to the area they are best suited to. Once this is complete, the requirements for the MVP are confirmed, along with a realistic schedule for development, testing, and rollout.

2. Prototype

The team builds a basic working version of the core intake-lookup workflow to demonstrate to stakeholders, gathering feedback that may adjust the scope or priority of remaining features.

3. Test and Refine

The prototype undergoes structured testing to identify and resolve bugs, confirm barcode scanning reliability, verify notification delivery, and ensure the system handles unexpected or invalid input gracefully.

4. Deploy

The system is rolled out in a controlled way, typically alongside the existing process for a short overlap period, with a small group of staff trained to use it before a full cutover.

5. Presentation and Sign-Off

The completed system is presented to stakeholders, who confirm that it meets the requirements set out in this document before it is formally signed off.

Project Execution

To keep the team coordinated and on schedule, the following practices will be used throughout the project:

Weekly Meetings

Short weekly check-ins ensure everyone is aligned on what is expected for that stage of the project and surface any blockers early.

Communication Channels

A primary channel (e.g. Discord or Microsoft Teams) is used for general project communication, supported by dedicated group chats for each core team.

Shared Timetable

A shared timetable of upcoming deliverables and deadlines is maintained and circulated across the team's communication channels.

Minimum Viable Product (MVP) Features

The MVP represents the core version of the Platform that delivers immediate operational value within the available project timeline, focused on replacing the manual intake, lookup, notification, and reporting workflow currently run through SharePoint.

1. Role-Based User Login and Dashboards

The Platform provides secure login for staff, with administrators able to access additional configuration and reporting tools not shown to front-desk staff. Each dashboard only displays functionality appropriate to that user's role.

2. Package Intake and Barcode Scanning

Staff log a new package on arrival by scanning its barcode (via dedicated scanner or webcam) or entering its details manually where no barcode exists. The system automatically records the date received and applies the flat collection fee, replacing the current manual SharePoint entry step.

3. Package Lookup, Status Update and Storage Tracking

Staff can search for a package by tracking number, customer name, or customer phone number, and update its status (for example: Received, Ready for Collection, Collected). This replaces the current manual search process.

4. Automatic Customer Notifications

When a package's status changes to 'Ready for Collection', the system automatically sends the customer an SMS or WhatsApp message, removing the need for staff to contact customers individually.

5. Staff Dashboard: Collected vs Uncollected

A live dashboard shows, at minimum: the total number of packages currently awaiting collection, the total collected within a selected period, and a simple visual comparison of collected versus uncollected packages, giving staff and management the throughput visibility that does not currently exist.

6. Income Reporting

A reporting module calculates total income collected (flat rate multiplied by packages collected) for any selected date range — for example, this week, this month, or a custom range — replacing the current manual receipt-reconciliation process entirely.

Recommended Technology Stack

Layer	Recommended Technologies	Description
Frontend	React, Vue, Angular	React is widely adopted and well suited to building a modular dashboard-driven interface, with a gentle learning curve and a large ecosystem of supporting libraries. Vue.js offers simplicity and readability, helpful for fast onboarding. Angular is powerful and full-featured, suitable where a more structured framework is preferred, though it has a steeper learning curve.
Backend	Node.js with Express, Laravel	Node.js with Express is ideal for projects needing quick development using JavaScript across the stack, and integrates well with SMS/WhatsApp messaging APIs. Laravel is well suited to teams more familiar with PHP, offering strong security and an elegant syntax.
Database	MySQL, Firestore	MySQL is reliable, familiar, and well suited to the structured, relational package and customer data used in this project. Firestore, a NoSQL database, offers more flexibility for unstructured data and may suit fast iteration cycles.
Version Control	GitHub	Provides robust version control, along with continuous integration (CI) pipelines and automated testing for the development team.
Notifications	[To be confirmed — e.g. Twilio, WhatsApp Business API]	An SMS/WhatsApp gateway is required to support automatic customer notifications; see the note below regarding provider selection.

NOTE FOR LECTURER/STUDENTS: *Please confirm whether there is a preferred or already-approved SMS/WhatsApp gateway provider, or a budget ceiling for messaging costs, so that students can select and cost a specific provider rather than leaving this generic.*

Appendix

Appendix A: Solution Comparison

Criteria	Courier Services Management and Tracking Platform (Solution 1)	Current System (Solution 2)
User Friendly	Medium – modern workflow, requires initial mentorship/training	High – proven, staff already familiar with it
Budget Consciousness	High – low development cost, modest ongoing messaging/hosting cost	High in the short term – no new spend, but hidden cost in ongoing staff time
Implementation	Medium – custom-built to fit the exact workflow	High – already in place, nothing to implement
Time Consciousness	Medium – phased rollout across a project timeline	High – instantly available, no rollout needed
Adaptability	High – can evolve with future operational needs	Low – locked into SharePoint's list-based structure
End User Instruction	Medium – short training tailored to the new workflow	High – no new training required
Human Resources	Medium – requires a development team and mentorship	Low ongoing efficiency – frees up build effort but costs staff time indefinitely

Conclusion: The Platform is feasible provided mentorship, phased rollout, and adequate technical support are in place. The current system is stable and familiar but does not scale well and leaves every identified operational problem unresolved.

Appendix B: Project Team Structure

Role	Team Members
Project Manager	[Name]
Front-End Team	[Names]
Backend & DevOps Team	[Names]
Documentation & QA	[Names]

Appendix C: Project Phases

Phase	Description
1. Assembling the Team & Planning	Assign team members to roles, define MVP requirements, and create the project timeline.
2. Prototype Development	Build a basic version of the intake-and-lookup workflow to present to stakeholders for feedback.
3. Testing & Refinement	Conduct bug fixes, scanning-reliability testing, notification testing, and error handling checks.
4. Deployment	Implement the system in a controlled environment alongside the existing process, with selected users.
5. Presentation & Sign-Off	Present the final product to stakeholders and confirm approval.

Appendix D: Communication Plan

Method	Purpose
Primary Channel (e.g. Discord/Teams)	Main communication platform for project updates and announcements.
Team Group Chats	Quick coordination within specific teams (front-end, back-end, QA).
Weekly Meetings	Track progress, resolve issues, and adjust timelines.

Appendix E: Risk Register

Risk	Impact	Mitigation
Development team workload / availability	May delay development or reduce feature completeness	Prioritise a minimal, well-scoped MVP; build in schedule buffers

Barcode scanning reliability	Could slow adoption if intake is unreliable	Test thoroughly with both scanning methods before go-live
Underestimated messaging volume	Ongoing notification costs could exceed budget	Confirm real parcel volume before finalising provider and budget
Data migration from SharePoint	Risk of data loss or duplication during cutover	Run both systems in parallel briefly before full cutover
Staff adoption resistance	Slower transition away from familiar manual process	Short, hands-on training; clear communication of time savings

Appendix F: Assumptions and Information Still Required

This document has been prepared as a detailed working template. The following items should still be confirmed to finalise the business case with real figures:

- Current parcel volume (per day/week/month).
- The flat collection fee amount, and whether it has changed historically or is expected to change.
- Full list of fields currently captured per parcel, if any differ from those assumed in this document.
- Whether barcode scanners are already owned, and whether staff workstations have webcams.
- Any preferred/approved SMS or WhatsApp provider, or a messaging budget ceiling.
- Overall budget ceiling and target delivery timeline for the project.
- Number of staff/workstations that will use the system, and whether a separate admin/manager role is required.